Balancing immunosuppression

Induction vs. maintenance

Complications

TRANSPLANT MEDICINE

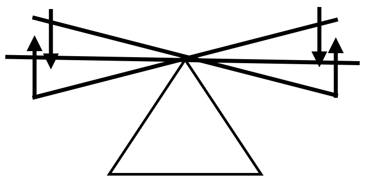
One of the core principles of transplant medicine is balancing immunosuppression.

- Oversuppression (*click*) of the immune system leads to infection/cancer.
- Undersuppression (*click*) puts patients at risk of organ rejection.

The likelihood of an organ to reject directly correlates with the degree of immunosuppression that a patient will require long term. Lung transplants are the most likely reject. Fifty percent of lung transplant patients experience rejection in 5 years. Liver transplants are least likely to reject.

Click on the "Oversuppression" and "Undersuppression" buttons. Then click anywhere to continue.

Immunosuppression



Rejection ⟷ Infection Cancer

Degree of immunosuppression

Lung > Heart/ Kidney > Liver

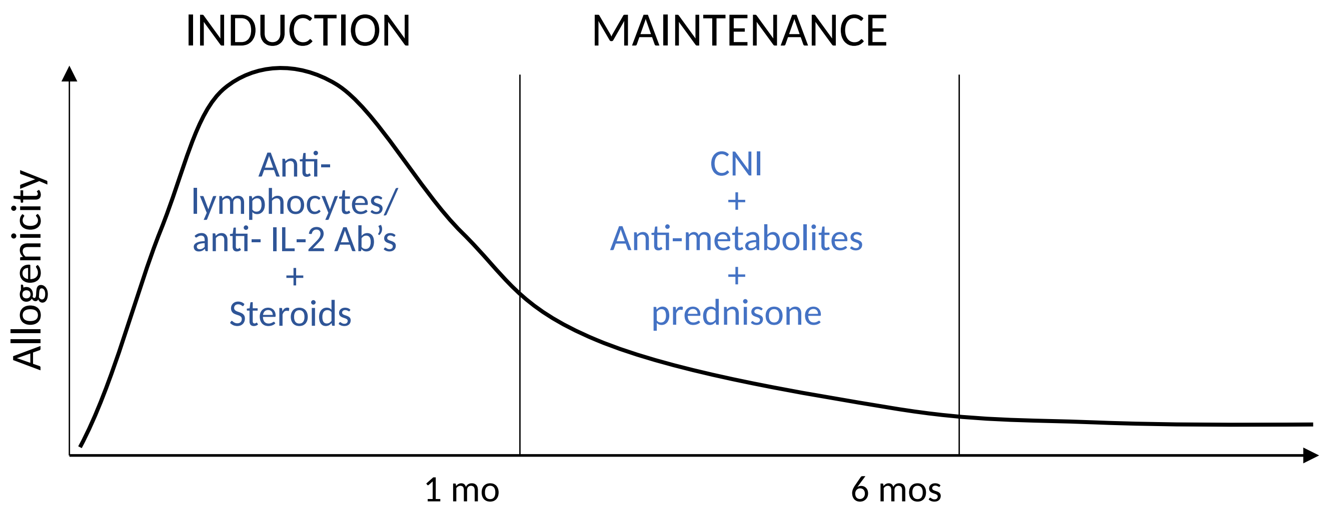
Balancing immunosuppression

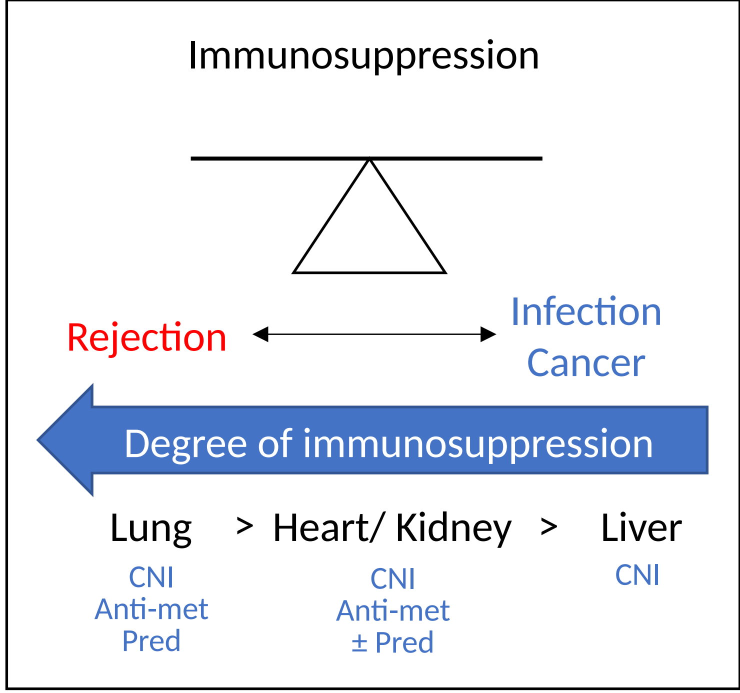
Induction vs. maintenance

Complications

TRANSPLANT MEDICINE

MEDICATIONS





The “allogenicity” of a transplant can be thought of as the likelihood of rejection. Immediately after transplant this is very high.

- Induction (*click*) immunosuppression is used immediately after a transplant. This typically consists of anti-lymphocyte antibodies (e.g., anti-thymocyte globulin) or anti-IL-2 antibodies in addition to high dose steroids. The exact regimen depends on the transplant.
- Maintenance (*click*) immunosuppression consists predominantly of 3 med classes.
  - Calcineurin inhibitors – tacrolimus, cyclosporine
  - Anti-metabolites – azathioprine, mycophenolate
  - Steroids – predominantly prednisone
  - Other classes include mTOR inhibitors (everolimus, sirolimus)

The likelihood of rejection decreases over time and the degree of immunosuppression can be decreased accordingly.

- Because lung transplants are most likely to reject, these patients are typically on all 3 classes of immunosuppressants even years after a transplant.
- Heart/kidney transplant patients will typically be on a CNI, anti-metabolite, and sometimes a steroid.
- Liver transplant patients may only be on a CNI several years after transplant.

Balancing immunosuppression

Induction vs. maintenance

Complications

TRANSPLANT MEDICINE

MEDICATIONS

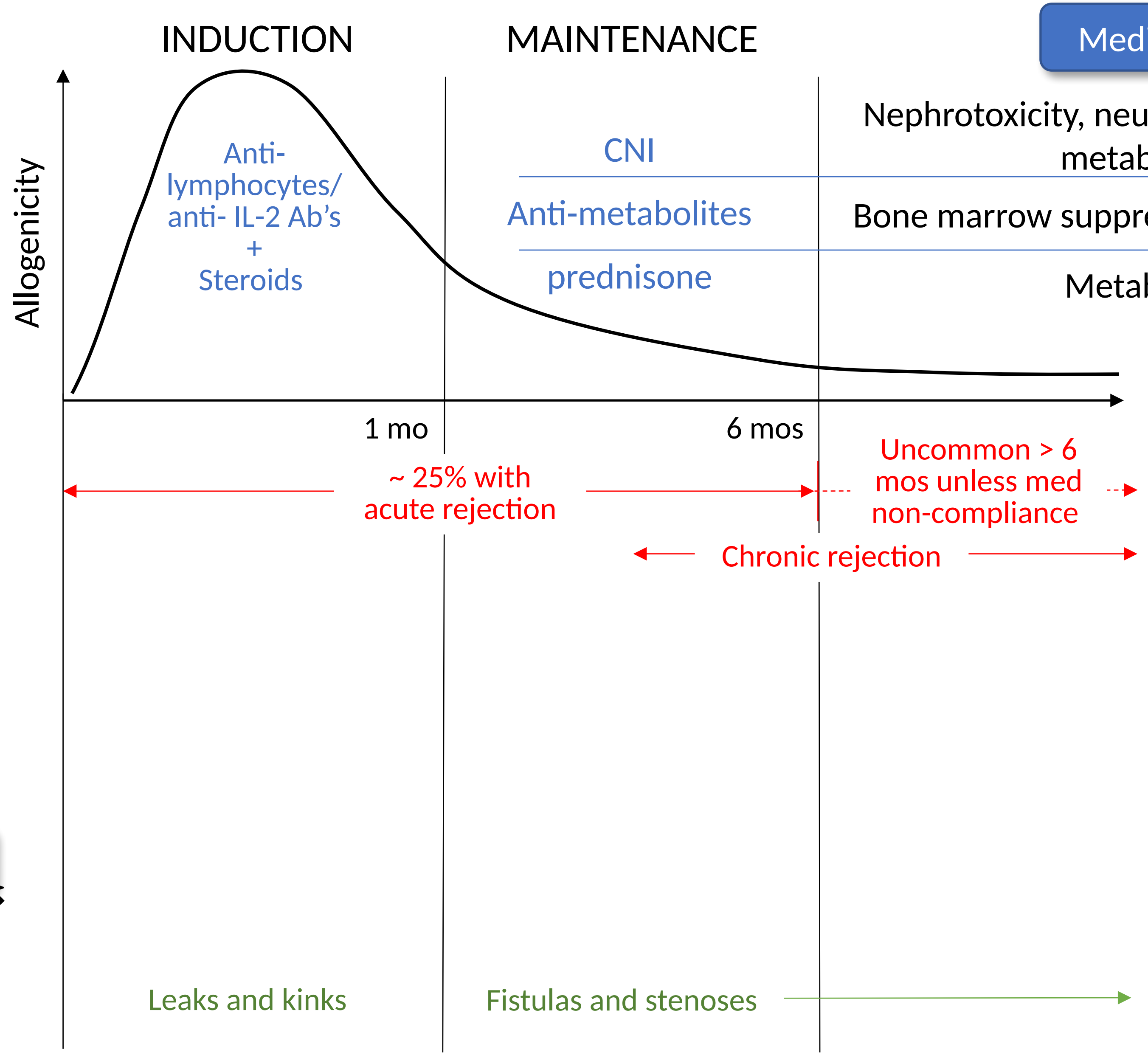
COMPLICATIONS

Rejection

Infections

Malignancy

Surgical



Medication Effects

Nephrotoxicity, neurotoxicity, ↑ K and uric acid, metabolic syndrome

Bone marrow suppression, GI side effects (colitis)

Metabolic syndrome

Rejection can be divided into acute rejection and chronic rejection.

- Acute rejection is further separated into cellular and humoral rejection.
    - Acute cellular rejection is T-cell mediated and is diagnosed on biopsy.
    - Acute humoral rejection is antibody mediated and can be diagnosed on biopsy or by presence of donor-specific antibodies.
- Chronic rejection can occur in certain organ transplants (lung and kidney are common).

Balancing immunosuppression

Induction vs. maintenance

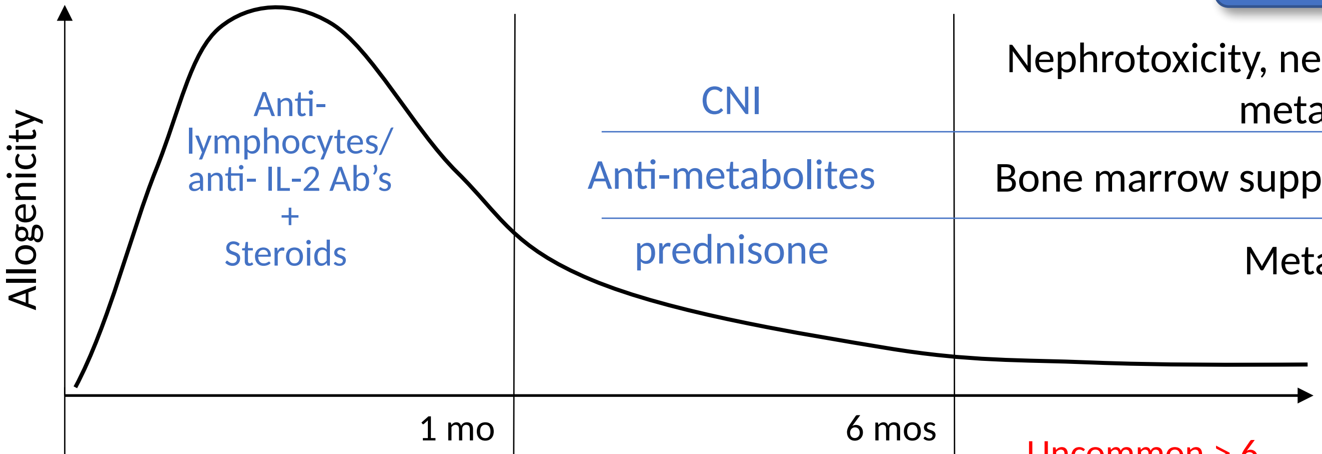
Complications

TRANSPLANT MEDICINE

Medication Effects

INDUCTION | MAINTENANCE

MEDICATIONS



- CNI: Nephrotoxicity, neurotoxicity, ↑ K and uric acid, metabolic syndrome
- Anti-metabolites: Bone marrow suppression, GI side effects (colitis)
- prednisone: Metabolic syndrome

COMPLICATIONS

Rejection

~ 25% with acute rejection

Uncommon > 6 mos unless med non-compliance

Chronic rejection

Infections

Donor/ recipient + Nosocomial

Reactivation (TB, HBV, HSV/VZV, CMV, BK) + OIs

Community acquired >> OIs (molds, Mucor, late viral)

Malignancy

Surgical

Leaks and kinks

Fistulas and stenoses

Infections can be divided by time from transplant and is a result of varying immunosuppression.

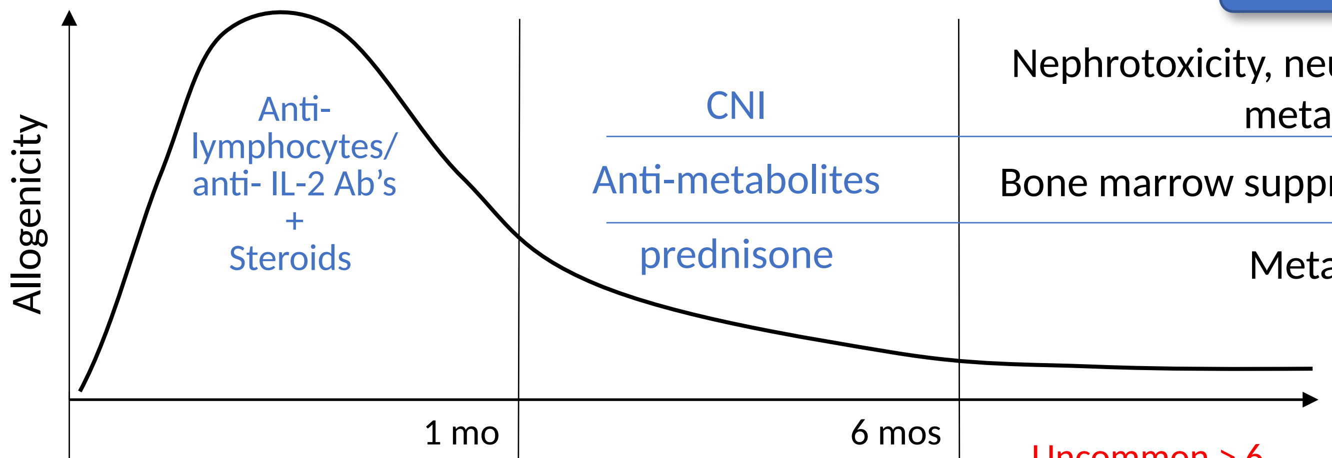
- <u><1 mo</u>: Nosocomial infections are common given hospitalization. Recipient derived infections are related to colonization by organisms (*Aspergillus*, *Pseudomonas*) that become pathogenic in the setting of immunosuppression. Donor-derived are uncommon (HSV, West nile)
- <u>1-6 mos</u>: Opportunistic infections such as PJP, Cryptococcus, CMV, toxoplasmosis are common. The incidence is reduced with prophylaxis.
- <u>>6 mos</u>: Community acquired infections and late viral infections (HSV, HBV, HCV, CMV).

Balancing immunosuppression

Induction vs. maintenance

Complications

TRANSPLANT MEDICINE



MEDICATIONS

INDUCTION	MAINTENANCE	Medication Effects
Anti-lymphocytes/ anti- IL-2 Ab's + Steroids	CNI	Nephrotoxicity, neurotoxicity, ↑ K and uric acid, metabolic syndrome
	Anti-metabolites	Bone marrow suppression, GI side effects (colitis)
	prednisone	Metabolic syndrome

COMPLICATIONS

	< 1 mo	1 mo – 6 mos	> 6 mos
Rejection	~ 25% with acute rejection	~ 25% with acute rejection	Uncommon > 6 mos unless med non-compliance
		Chronic rejection	Chronic rejection
Infections	Donor/ recipient + Nosocomial	Reactivation (TB, HBV, HSV/VZV, CMV, BK) + OIs	Community acquired >> OIs (molds, Mucor, late viral)
Malignancy	Overall 2x as likely.	PTLD (multiorgan/ SB > lung > kidney/heart > liver)	Skin cancer, anogenital
Surgical	Leaks and kinks	Fistulas and stenoses	→

Overall, the risk of malignancy in transplant patients is 2-fold higher than the general population. They are at particular risk of virally mediated cancers (lymphomas, liver cancer, skin and anogenital cancers). Most of these cancer appear late (after several years of immunosuppression) except for posttransplant lymphoproliferative disorder (PTLD). PTLD typically occurs in the first year. EBV positivity and degree of immunosuppression are risk factors for PTLD.